



Anesthetic complications

Systemic complications of LA

- Over Dose Reaction (LA Toxicity)
- Allergic Reactions

LOCAL ANAESTHETIC TOXICITY (over dose)

It is the result of high concentration of the LA drug in the blood stream to adversely affect CNS, Respiratory system and/or the circulatory system.

This may be due to:

- 1- Too large dose of LA drug.
- 2- Accidental intravenous injection of the drug.
- 3- Use of highly toxic drug.
- 4- Injection into highly vascular area without addition of vasoconstrictor.
- 5- Unusually slow detoxification or elimination as a result of severe liver or kidney dysfunction .

Predisposing factors for over dose

Patient Factor

- Age
- Weight
- Other Medication
- Presence of Disease
- Genetics

Drug Factor

- Vasoactivity
- Concentration of drug
- Dose
- Route of administration
- Rate of injection

The early symptoms of toxic overdose are primarily those of C.N.S. stimulation . The pt. becomes talkative, apprehensive, excited and often complains of sudden headache.

An increase in pulse , BP, and RR may occur in advanced part of stimulation phase.

Tremors and twisting of small muscles like muscles of face and fingers.

At times convulsions may occur.

Pt may vomit and feels nauseated.

Con.



C.N.S. stimulation is usually followed by a proportionate degree of C.N.S. depression leading to drop in BP, weak and rapid pulse and sometimes apnea.

Unconsciousness may and usually does resulting from severe C.N.S. depression and death may follow because of hypoxia and its effect on cardiac system.



Once the symptoms of toxic overdose appears, they should be recognized and treated. This means the pt. must be watched closely during the injection and for a reasonable time after injection

We have to know the sooner the symptoms are recognized and ttt instituted, the more certain will be the chance of a favorable result.

Treatment of Toxicity

Mild LA overdose reactions are managed by monitoring vital signs, administer Oxygen and observe the pt in the office for 1 hour before leave.

If signs of anesthetic toxicity do not rapidly disappear, a slow IV 2.5-5mg Diazepam should be given. Medical assistance should be called if signs do not resolve or progressively worsen.

Cont.

In case of severe reaction (seizure, cardiac dysrhythmia or arrest) The following should be done quickly:

- 1- the pt should be placed in supine position.
- 2- if convulsion, protect the pts from hurting themselves.
- 3- Call medical assistance.
- 4- Monitor vital signs & administrate oxygen.
- 5- IV Diazepam 5-10mg.
- 6- Institute basic life support if necessary .

Importance of prevention

It is necessary to emphasize the importance of prevention rather than ttt by the following :

- 1- Adequate pre-anesthetic evaluation of the pt.
- 2- The least possible volume should be used.
- 3- The weakest concentration compatible with successful anesthesia should be used.
- 4- The injection should be made slowly and always aspirate before injection.

Allergic & Anaphylactic reactions

It is a specific antigen-antibody reaction and the pt should has been previously sensitized to a particular drug or a chemical derivation of the drug.

The reactions may manifest as

- 1- Rashes 2- Urticaria 3- Rhinitis
- 4- Angioneutotic edema 5- Asthma

Allergic reactions may occur immediate or delayed after drug administration

Management of Allergic reaction

1. Benadryl (anti-histamine) 20-40mg IV or IM.
2. Epinephrine 1/1000 conc. 0.2-0.5 IM or Subcu.
3. Aminophylline IV when tracheobronchial tree is involved.
4. Monitor vital signs frequently.

Anaphylactic reaction is one of medical emergencies in the dental office. It is form of allergy associated with severe hypotension and loss of consciousness. It can statrt in 5-30 min .The pulse will be very week and pt becomes cyanotic with difficult respiration.

It is true emergency that might lead to death unless prompt action done

Mangement of anaphylactic reaction

1. The pt should be placed immediately in a supine position with the leg elevated.
2. Respiration should be supported with 100% oxygen.
3. Adrenaline 0.3-0.5ml 1/1000 should be given Subcut. Or IM and can be repeated in case of need.
4. Antihistamine IV OR IM .
5. Hydrocortisone 200 mg can be given .
6. Basic life support and monitor the vital sign

the most commonest complications related to LA administration or to the dental visit

Fainting or Syncope(Vaso vagal attack)

It is the most common emergency in the dental office. It is due to temporary cerebral anemia.

The pt appears pale, dizzy, nauseated with cold and sweaty skin but still in harmony with the surroundings.

Loss of consciousness may follow as an extreme manifestation of cerebral anemia.

The fainting spell can be stopped by improving the cerebral circulation by lowering the head and elevation of the legs. This can be done quickly in the dental office by tilting the dental chair

A diffusible stimulant such as aromatic spirits of ammonia can be held under the nose for several breaths.

